G18 Project 2: UNIX-like Mini Shell and Utilities

1 Project Overview

This project implements a lightweight UNIX-like shell and a set of custom file/text utilities in C.

The shell prompt is:

```
g18-shell>
```

Each command runs as an external executable from the `bin/` directory.

2 Objectives

1. Build a working shell that accepts user commands and executes utilities.
2. Implement common UNIX-style commands from scratch in C.
3. Keep the build process simple and reproducible using `make`.
4. Maintain a clean source layout for team development.

3 Build and Run

3.1 Build all binaries

```
make
```

3.2 Run shell

```
./bin/g18-shell
```

3.3 Clean build outputs

```
make clean
```

Notes:

1. `make` auto-compiles every `.c` file inside `src/utils/` into `bin/`.
2. If old binaries are present, run `make clean` before rebuilding.

4 Project Structure

```
G18_Project2_1/
|-- Makefile
|-- README.md
|-- include/
|   `-- common.h
|-- src/
|   |-- shell/
|   |   `-- main_shell.c
|   `-- utils/
|       |-- g18_cat.c
|       |-- g18_cp.c
|       |-- g18_grep.c
|       |-- g18_ls.c
|       |-- g18_mkdir.c
|       |-- g18_mv.c
|       |-- g18_rm.c
|       |-- g18_touch.c
|       `-- g18_wc.c
|-- bin/                  # Generated binaries
`-- tests/
```

5 Shell Design

File: `src/shell/main_shell.c`
Features:

1. Shows prompt `g18-shell>`.
2. Reads one command line from `stdin`.
3. Splits command and arguments by spaces.
4. Supports built-in `exit`.
5. Uses `fork()` + `execv()` to run utilities from `./bin/<command>`.
6. Parent process waits for child completion using `wait()`.

6 Implemented Utilities

All utilities are implemented and available in `src/utils/`.

6.1 g18_ls

File: `src/utils/g18_ls.c`
Purpose: The `ls` utility is used to list directory contents.

Features:

- Includes hidden files with `-a` option.
- Shows long listing format with `-l` option.

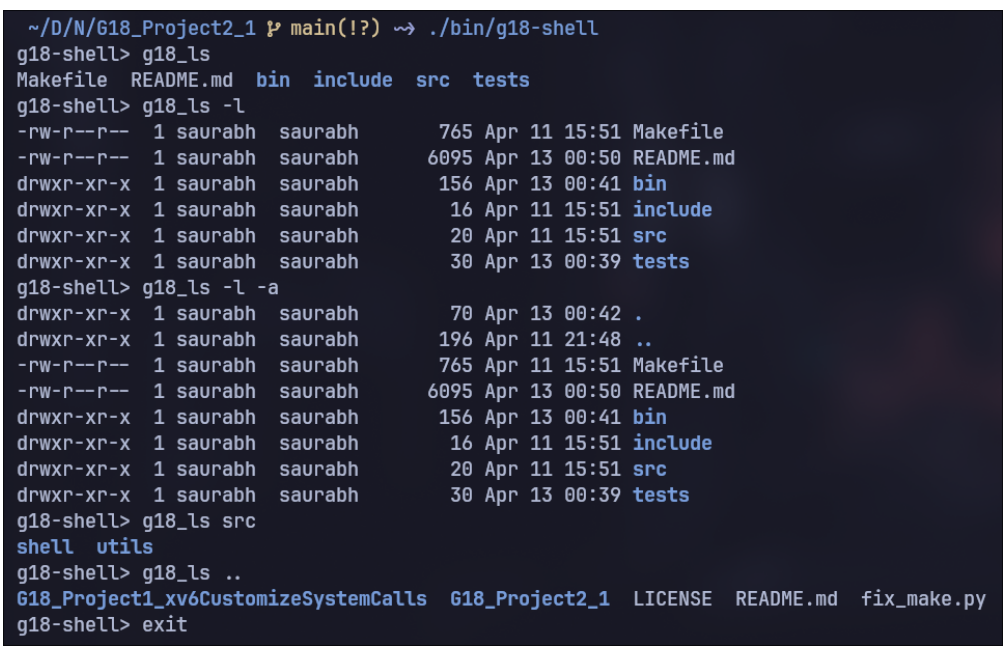
Usage:

```
g18_ls [-a] [-l] [directory]
```

Implementation Details:

- Uses `scandir()` with `alphasort` for ordered output.
- Uses `lstat()` for metadata in long mode.
- Shows colored directory names.

Screenshot:



6.2 g18_cat

File: `src/utils/g18_cat.c`
Purpose: The `cat` utility is used to display the contents of a file. It can also optionally display line numbers for each line.

Features:

- Displays contents of one or more files.
- Supports optional argument `-n` to show line numbers.
- Reads from standard input if no file is provided.

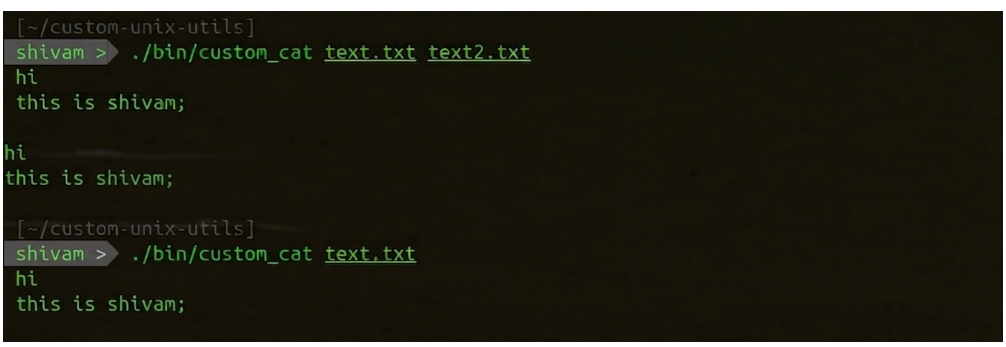
Usage:

```
g18_cat file.txt
g18_cat -n file.txt
g18_cat file1.txt file2.txt
```

Implementation Details:

- Uses `fgets()` for line-by-line reading when line numbering is enabled.
- Uses `fread()` and `fwrite()` for efficient block-based copying otherwise.
- Supports multiple files by iterating through command-line arguments.
- Falls back to standard input if no file is specified.

Screenshot:



6.3 g18_wc

File: `src/utils/g18_wc.c`
Purpose: The `wc` utility is used to count lines, words, and characters in a specified file.

Features:

- Shows line count only with `-l` option.
- Shows word count only with `-w` option.

- Shows character count only with `-c` option.

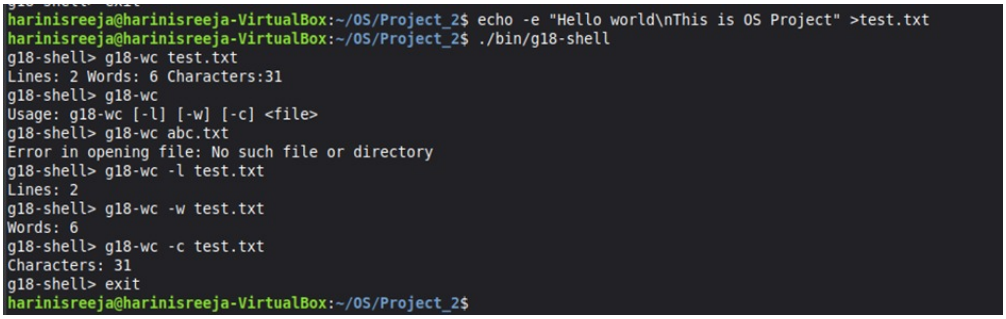
Usage:

```
g18_wc [-l] [-w] [-c] <file>
```

Implementation Details:

- If no flag is given, prints all three counts.
- Uses simple whitespace boundary logic for words.

Screenshot:



6.4 g18_touch

File: `src/utils/g18_touch.c`
Purpose: The `touch` utility is used to create a new file if it does not exist or update the access and modification timestamps of an existing file.

Features:

- Creates files if they do not exist.
- Updates file timestamps using system calls.

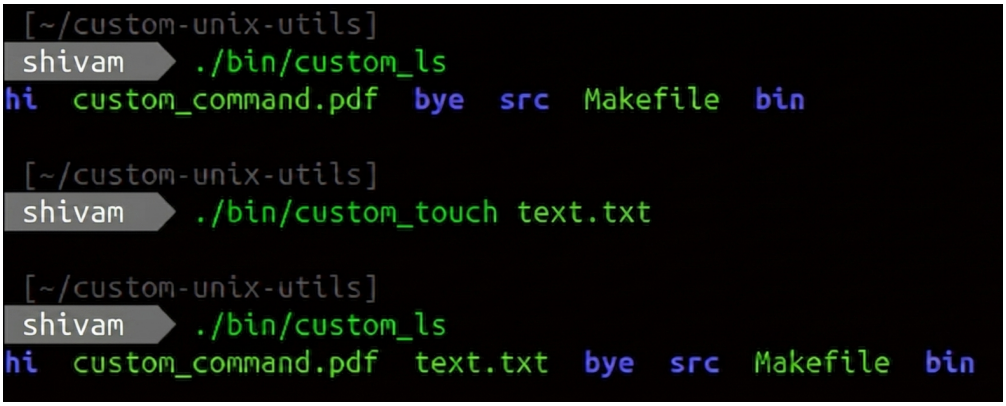
Usage:

```
g18_touch file.txt
g18_touch file1.txt file2.txt
```

Implementation Details:

- Uses `open()` with `O_CREAT` flag to create files if they do not exist.
- Uses `utime()` to update file timestamps.
- Iterates through all provided file arguments.
- Handles errors using `perror()`.

Screenshot:



6.5 g18_cp

File: `src/utils/g18_cp.c`
Purpose: The `cp` utility is used to copy a source file to a destination file.

Features:

- Copies file contents byte by byte.

Usage:

```
g18_cp <source_file> <destination_file>
```

Implementation Details:

- Opens source in read mode and destination in write mode.
- Copies data byte by byte using `fgetc()` and `fputc()`.

Screenshot:

```
shalvi@shalvi-VirtualBox:~$ mkdir unix_utilities
shalvi@shalvi-VirtualBox:~$ cd unix_utilities/
shalvi@shalvi-VirtualBox:~/unix_utilities$ mkdir src bin include
shalvi@shalvi-VirtualBox:~/unix_utilities$ ls
bin  include  src
shalvi@shalvi-VirtualBox:~/unix_utilities$ cd src/
shalvi@shalvi-VirtualBox:~/unix_utilities/src$ nano g18_cp.c
shalvi@shalvi-VirtualBox:~/unix_utilities/src$ cd ..
shalvi@shalvi-VirtualBox:~/unix_utilities$ nano Makefile
shalvi@shalvi-VirtualBox:~/unix_utilities$ make
make: *** No rule to make target 'src/custom_cp.c', needed by 'custom_cp'.  Stop
.
shalvi@shalvi-VirtualBox:~/unix_utilities$ nano g18_cp.c
shalvi@shalvi-VirtualBox:~/unix_utilities$ rm g18_cp.c
rm: cannot remove 'g18_cp.c': No such file or directory
shalvi@shalvi-VirtualBox:~/unix_utilities$ nano Makefile
shalvi@shalvi-VirtualBox:~/unix_utilities$ make
gcc -Wall src/g18_cp.c -o bin/g18_cp
shalvi@shalvi-VirtualBox:~/unix_utilities$ ls bin
g18_cp
shalvi@shalvi-VirtualBox:~/unix_utilities$ echo "project-2 OS" > file1.txt
shalvi@shalvi-VirtualBox:~/unix_utilities$ cat file1.txt
project-2 OS
shalvi@shalvi-VirtualBox:~/unix_utilities$ ./bin/g18_cp file1.txt file2.txt
```

```
shalvi@shalvi-VirtualBox:~/unix_utilities$ ./bin/g18_cp file1.txt file2.txt
File copied successfully!
shalvi@shalvi-VirtualBox:~/unix_utilities$ cat file2.c
cat: file2.c: No such file or directory
shalvi@shalvi-VirtualBox:~/unix_utilities$ cat file2.txt
project-2 OS
shalvi@shalvi-VirtualBox:~/unix_utilities$
```

6.6 g18_mv

File: `src/utils/g18_mv.c`
Purpose: The `mv` utility is used to move or rename a file.

Features:

- Moves or renames the specified file to the destination path.

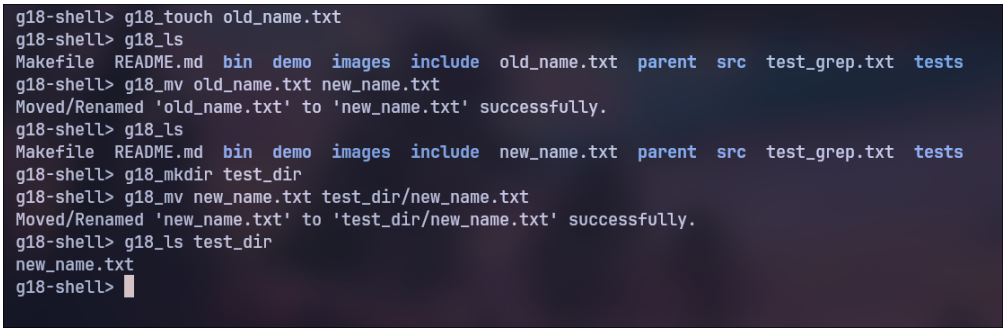
Usage:

```
g18_mv <source> <destination>
```

Implementation Details:

- Uses `rename()` for the move/rename operation.

Screenshot:



6.7 g18_rm

File: `src/utils/g18_rm.c`
Purpose: The `rm` utility is used to remove files and directories.

Features:

- Allows directory removal with `-r` option.

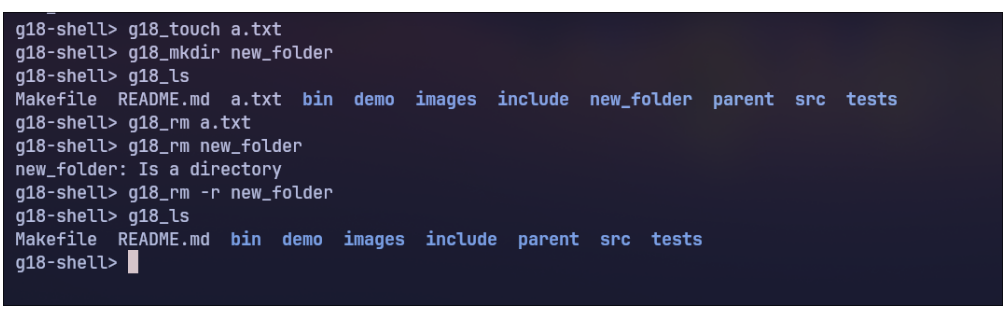
Usage:

```
g18_rm [-r] <path...>
```

Implementation Details:

- Tries `unlink()` first.
- If `-r` is set, attempts `rmdir()` for directory paths.

Screenshot:



6.8 g18_mkdir

File: `src/utils/g18_mkdir.c`
Purpose: The `mkdir` utility is used to create one or more directories.

Features:

- Creates parent directories as needed with `-p` option.

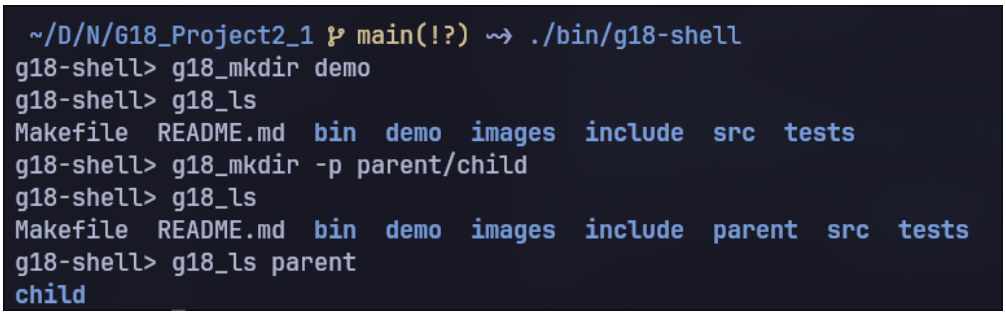
Usage:

```
g18_mkdir [-p] <directory...>
```

Implementation Details:

- Normal mode uses `mkdir(path, 0777)`.
- `-p` mode creates intermediate directories safely.

Screenshot:



6.9 g18_grep

File: `src/utils/g18_grep.c`
Purpose: The `grep` utility is used to search for a pattern in one or more files.

Features:

- Supports case-insensitive search with `-i` option.
- Prints line numbers with `-n` option.
- Prints match count only with `-c` option.
- Inverts match with `-v` option.

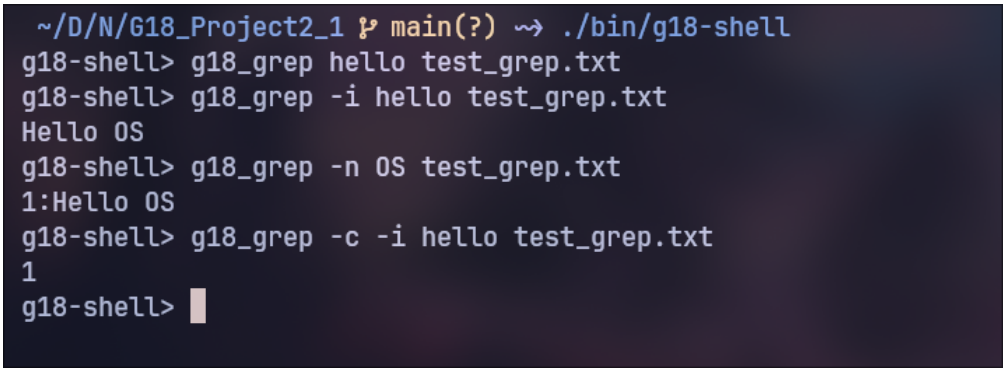
Usage:

```
g18_grep [-i] [-n] [-c] [-v] <pattern> <file...>
```

Implementation Details:

- Supports single-file and multi-file output formatting.
- Returns exit code `0` if at least one match is found, otherwise `1`.

Screenshot:



7 Error Handling Approach

1. Invalid usage prints a usage message.
2. File and directory failures report system errors (`perror` or equivalent messages).
3. Shell prints a clear message if command is not found in `./bin/`.

8 Project Team Contributions

The following details the specific contributions for Project 2:

- **24JE0689 - Satyavarapu Surya Sitha Harini Sreeja**: Implemented the `g18_wc` utility for line, word, and byte counting.
- **24JE0690 - Saurabh Prajapat**: Set up the Project 2 architecture and implemented the `g18_ls` (with `-a`, `-l` flags) and `g18_mkdir` (with `-p` flag) utilities.
- **24JE0691 - Shaik Zaheer Mushtaq**: Implemented the `g18_grep` utility with advanced pattern matching flags (`-i`, `-n`, `-c`, `-v`) and standard input support.
- **24JE0692 - Shalvi Kumari**: Implemented the `g18_cp` utility handling safe source-to-destination file copy operations and error checking.
- **24JE0693 - Shivam Chaudhary**: Implemented `g18_touch` for file creation and `g18_cat` with standard output handling and line numbering (`-n`).
- **24JE0694 - Shlok Radadia**: Implemented `g18_rm` (with continuous and recursive `-r` file deletion) and `g18_mv` for move/rename operations.